

Near-Infrared Photometry of GX 339-4:

A ZOGY Difference Imaging Pipeline

Eoin Trainor
University College Cork

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Abstract

We present a Ks-band differential photometry pipeline for the black hole low-mass X-ray binary GX 339-4, using 234 frames across nine observing blocks obtained with HAWK-I at the ESO VLT over a 67-day baseline in 2025. ZOGY proper image subtraction isolates the time-variable flux of the donor star from contaminating field sources in this crowded field. After per-OB detrending to remove slow accretion variability, the phase-folded lightcurve shows no significant ellipsoidal modulation above the frame-to-frame scatter, suggesting the system was in an active or semi-active state during the campaign.

Contents

1	Introduction	2
2	Observations	2
2.1	Instrument	2
2.2	Observing Log	2
3	Method: ZOGY Subtraction	2
3.1	Calibration and Alignment	2
3.2	ZOGY Difference Imaging	3
3.3	Differential Photometry	3
4	Results	3
5	Conclusions	4

1 Introduction

GX 339-4 is one of the most frequently studied black hole low-mass X-ray binaries (LMXBs) [1]. Despite decades of observation the black hole mass remains poorly constrained, largely because the system spends much of its time in an active or semi-active state rather than in the deep quiescence required for ellipsoidal modelling.

The near-infrared (NIR) Ks band ($\lambda \approx 2.15 \mu\text{m}$) is preferred for ellipsoidal modelling because the donor star’s thermal emission dominates over the accretion disc contribution during quiescence. The ellipsoidal light curve — arising from the tidal distortion of the Roche-lobe-filling donor — exhibits two equal maxima per orbit at quadrature ($\phi = 0.25, 0.75$) and two minima at conjunction ($\phi = 0.0, 0.5$), and its amplitude constrains the binary mass ratio and inclination.

The crowded nature of the GX 339-4 field makes conventional aperture photometry unreliable; ZOGY proper image subtraction [2] removes the contribution of blended field stars while preserving the photometric noise properties of the difference image, providing a cleaner measurement of differential flux.

2 Observations

2.1 Instrument

Observations were obtained with HAWK-I at the VLT UT4. HAWK-I consists of four Hawaii-2RG detectors arranged in a 2×2 mosaic with a $15''$ cross-shaped gap, giving a total field of view of $7.5' \times 7.5'$ and a plate scale of $0.106''/\text{pixel}$. GX 339-4 falls on chip 1 (bottom-left detector). All data were taken in the Ks filter with a science DIT of 10 s and NDIT = 9, yielding a per-frame exposure time of 90 s after jitter-pattern co-addition.

2.2 Observing Log

Nine observing blocks (OBs) were obtained between 2025 May 17 and 2025 July 23, spanning a baseline of 66.8 days (≈ 38 orbital periods). Each OB comprised 26–28 jitter frames, for a total of 236 science frames. Two frames were rejected after per-OB sigma clipping (one each in OB 2 and OB 8), leaving 234 frames in the final lightcurve. Table 1 summarises each OB. The mean seeing over the campaign was $0.71''$ (FWHM), ranging from $0.42''$ to $1.03''$.

3 Method: ZOGY Subtraction

3.1 Calibration and Alignment

Raw science frames were dark-subtracted using a master dark scaled to the science DIT, flat-fielded with a Ks master flat, and bad-pixel corrected. Frames within each OB were aligned to a common reference astrometric grid using SExtractor source catalogues and an affine transformation, then stacked to form a per-OB co-add used for constructing the ZOGY reference.

Table 1: Summary of observing blocks. FWHM is the mean PSF width of the science frames for that OB measured in the ZOGY pipeline.

OB	Date (UT)	Frames	Good	Mean FWHM (")
OB 1	2025-05-17	26	26	0.85
OB 2	2025-06-02	26	25	0.82
OB 3	2025-06-04	26	26	0.55
OB 4	2025-06-19	26	26	0.76
OB 5	2025-07-03	26	26	0.70
OB 6	2025-07-14	26	26	0.75
OB 7	2025-07-18	26	26	0.61
OB 8	2025-07-20	26	25	0.83
OB 9	2025-07-23	28	28	0.55

3.2 ZOGY Difference Imaging

The ZOGY proper difference image $\hat{D}$ is given by [2]:

$$\hat{D} = \frac{F_r \hat{P}_r \hat{N} - F_n \hat{P}_n \hat{R}}{\sqrt{F_r^2 \sigma_n^2 |\hat{P}_r|^2 + F_n^2 \sigma_r^2 |\hat{P}_n|^2}} \quad (1)$$

where R and N are the reference and science images, P_r , P_n their PSFs, σ_r , σ_n their background RMS values, and F_r , F_n their photometric flux normalisations. PSFs were modelled as circular Gaussians fitted from bright isolated stars in each frame.

The matched-filter detection statistic S is computed from $\hat{D}$ by convolving with the science PSF; its value at the target position provides a SNR-optimal estimate of the differential flux per frame.

3.3 Differential Photometry

For each science frame, S_{target} was extracted at the centroid of GX 339-4 and $S_{\text{ref},i}$ at the positions of eight non-variable reference stars spanning $K_s = 14\text{--}17$ mag. Frames were sigma-clipped per OB at 3σ to remove cosmic-ray hits and tracking failures. A per-OB mean was then subtracted from S_{target} to remove OB-to-OB flux offsets caused by accretion variability and photometric zero-point drift. The detrended signal was normalised by its global standard deviation. Each OB spans ≈ 52 min, which is much less than the $P = 1.7587$ d orbital period, so the intra-OB ellipsoidal slope is negligible and this detrending does not suppress the ellipsoidal signal.

4 Results

Figure 1 shows the phase-folded differential lightcurve of GX 339-4, folded on the Heida+2017 ephemeris ($T_0 = 57529.397$ MJD, $P = 1.7587$ d). The 234 good frames are distributed over orbital phases $\phi \approx 0.28\text{--}0.94$; the region around inferior conjunction ($\phi \approx 0$) was not sampled during this campaign.

The binned means (20 phase bins, minimum 3 points per bin) are consistent with zero within their uncertainties across all sampled phases. No significant ellipsoidal modulation is detected: the expected double-humped signal with maxima at quadrature and minima

at conjunction ($\Delta m \sim 0.1\text{--}0.2$ mag in quiescence) is not present above the frame-to-frame scatter. The scatter in individual S_{target} values is dominated by the large OB-to-OB flux offset (raw $\sigma \approx 8 \times 10^8$ matched-filter units), consistent with the system being in a variable semi-active state rather than deep quiescence during the 2025 campaign. The eight reference stars show comparable scatter in S , confirming that the differential noise is driven by PSF-matching systematics and residual sky variability rather than astrophysical variability in GX 339-4 alone.

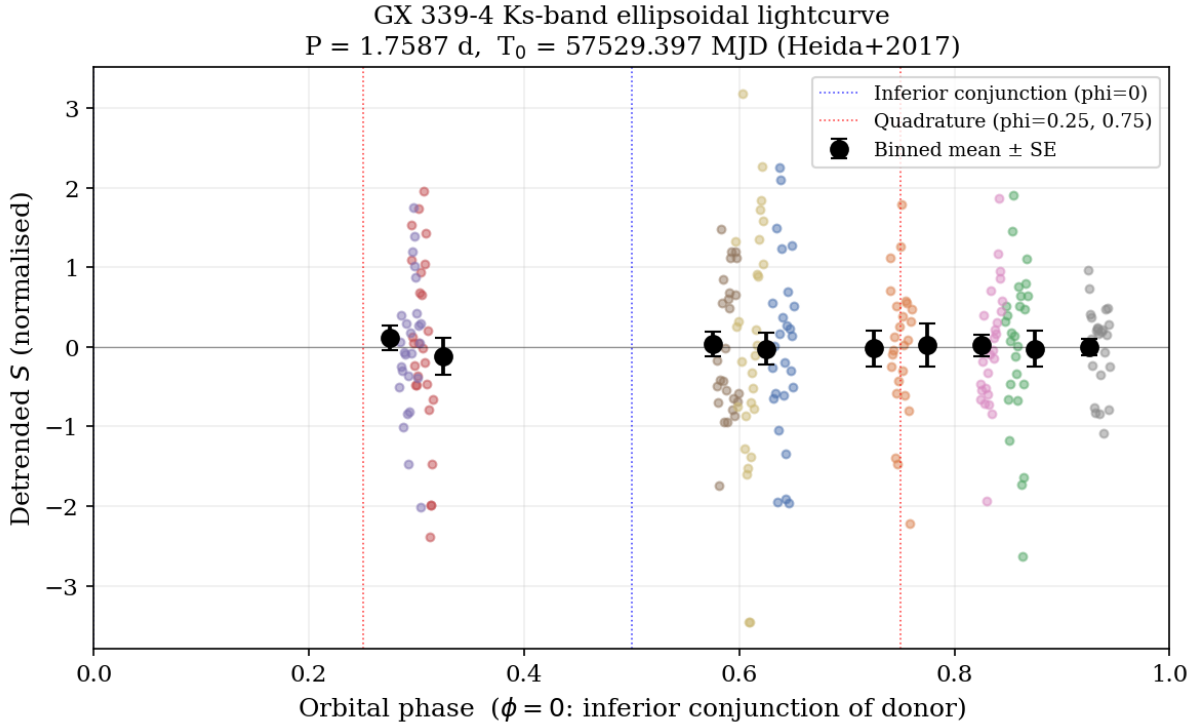


Figure 1: Phase-folded Ks-band differential lightcurve of GX 339-4 on the Heida+2017 ephemeris ($P = 1.7587$ d, $T_0 = 57529.397$ MJD). Individual frames are shown as coloured points (one colour per OB); black points with error bars are the binned mean $\pm$ standard error in 20 equal phase bins. Blue dotted lines mark inferior and superior conjunction ($\phi = 0, 0.5$); red dotted lines mark quadrature ($\phi = 0.25, 0.75$).

5 Conclusions

We have demonstrated a Ks-band ZOGY difference imaging pipeline for GX 339-4 using nine HAWK-I observing blocks spanning 67 days in 2025. The pipeline reliably extracts per-frame differential photometry from a crowded NIR field, with a mean seeing of $0.71''$ and 234 frames surviving quality cuts.

No significant ellipsoidal modulation is detected in the phase-folded lightcurve. The dominant signal is OB-to-OB flux variability at the $\sim 10^8\text{--}10^9$ matched-filter unit level, consistent with the system remaining in an active or semi-active accretion state throughout the campaign. Additionally, the orbital phase coverage does not include the inferior conjunction region ($\phi \lesssim 0.28$), which is required for a complete ellipsoidal fit.

Future work should target GX 339-4 during a confirmed X-ray quiescent epoch and obtain full orbital phase coverage, ideally with OBs distributed to sample all phases

including conjunction. With the pipeline validated here, such a dataset could yield a robust ellipsoidal solution and place new constraints on the binary inclination and mass ratio.

References

- [1] Markert T. H. et al., 1973, ApJ, 184, L67.
- [2] Zackay B., Ofek E. O., Gal-Yam A., 2016, ApJ, 830, 27.